

Diagnostics Module — End-to-End Test Plan

All roles, full booking → execution → verification → billing pipeline

Scope: Diagnostics module only — Pathology, Imaging, Outsourced Labs (OPD/Walk-in source). IPD/Emergency integration is a later build and is out of scope for this plan.

Environment: MBS Hospital tenant (dev/staging)

Tenant ID (x-tenant-id): acbda8d2-1730-4dd6-99a7-612b7559f07f / subdomain mbs

How to use this document: Work through each section top to bottom, in role order (a later role's worklist depends on an earlier role's action). Tick the checkbox for each step once verified, and note any deviation in the space provided.

Test Accounts

Role	Employee ID	Password	Used in
Doctor (OPD)	MBSDOC2026001	Test@12345	Section A
Reception	MBSRECP2026001	Test@12345	Sections B, I
Phlebotomist	MBSPHLEB001	Test@12345	Section C
Lab Technician	MBSLABTECH001	Test@12345	Sections D, H
Pathologist	MBSPATH001	Test@12345	Sections E, H
Radiographer	MBSRADIOG001	Test@12345	Section F
Radiologist	MBSRADIOL001 / MBSRADIOL002	Test@12345	Section G (two accounts needed for the separation-of-duties check)
Cashier	MBSCA2026001	Test@12345	Section I
Admin	ADMIN-001	Admin@12345 (+ MFA)	Section J
Super Admin	(super-admin portal login)	—	Section K

Precondition check before starting: confirm at least one active PATHOLOGY test (e.g. Complete Blood Count), one active IMAGING test with a mapped resource (machine/room), one active OUTSOURCED test, and a default Cash rate list with prices set for all three, already exist in the catalogue for this tenant. If not, complete Section J (Admin setup) first.

Section A — Doctor: Recommend Investigations

Login as **DOCTOR** MBSDOC2026001. Confirms a doctor can record a clinical recommendation during an OPD visit, distinct from actually booking/paying for it.

#	Action	Expected Result	✓
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A1	Open an OPD visit for a test patient, go to the Investigations / Recommended Investigations section.	Investigations picker loads with searchable catalogue tests (Pathology, Imaging, Outsourced all selectable).	☐
A2	Add two tests to the same recommendation action: one Pathology (e.g. CBC) and one Imaging test (e.g. X-Ray Chest).	Both save as separate PENDING recommendations linked to this visit and this doctor.	☐
A3	Add a free-text test not in the catalogue (type a name manually).	Recommendation saves with a "Not in catalogue" badge visible later at booking — cannot be booked until an admin adds it to the catalogue.	☐
A4	Save the visit/prescription.	No errors; recommendations are queryable by Reception for this patient.	☐

Section B — Reception: Book & Pay for Tests

Login as **RECEPTION** MBSRECP2026001. Covers the mixed-type single-order booking and the now-mandatory full-payment-at-booking rule.

#	Action	Expected Result	✓
B1	Open the patient's Book Test screen. Confirm both the Pathology and Imaging recommendations from Section A are listed as selectable, and the free-text one shows "Not in catalogue" (with a "Link to catalogue" option).	All three recommendations visible; only catalogue-linked ones are selectable checkboxes.	☐
B2	Select both the Pathology and Imaging tests together in one booking.	The page shows <i>both</i> a "Collection (Pathology / Outsourced)" section <i>and</i> a "Slot (Imaging)" section at once — no need to book them separately.	☐
B3	Fill in collection date/window/location for the Pathology item, and pick a machine/room + an available slot for the Imaging item.	Slot picker loads real availability for the chosen date; unavailable times are greyed out.	☐
B4	Observe the "Amount due" field before entering any payment.	It is read-only , pre-computed from the rate list (sum of both tests' prices) — there is no way to leave it blank or type a lower amount.	☐
B5	Try to click Book without selecting a payment method.	Blocked with "Select a payment method — full payment is required at booking."	☐
B6	Select a payment method (e.g. Cash) and click "Book & Collect ₹<amount>".	Order books successfully (HTTP 201); confirmation screen shows both tests with a PAID status.	☐
B7	From the sidebar, open the new "Diagnostic Orders" page (Reception-only, read-only).	The order just booked appears in the list with patient name, both test names, order status, and payment status — no edit/delete actions available.	☐
B8	Book a second, single-type order (Pathology only) for another patient, this time selecting "Home collection" if the test allows it.	Home collection option is enabled only when every selected collectable test supports it; booking succeeds the same way (full payment required).	☐
B9	Open Billing → Diagnostics tab (renamed from "Lab") for today.	Both freshly booked orders already appear as real bills — <i>immediately</i> , not after any lab work — each bill item tagged PATHOLOGY / RADIOLOGY / OUTSOURCED as appropriate.	☐
B10	Open the order detail page and attempt to cancel the now-paid order.	Blocked with "Cannot cancel an order once payment has been collected — there is no refund. If a sample needs to be redone, use the reject-and-recollect action on the affected item instead."	☐
B11	Open the "Pending Follow-Up" quick link on the Diagnostics dashboard.	The still-unbooked free-text recommendation from Section A appears once it crosses the configured pending-days threshold.	☐

Section C — Phlebotomist: Sample Collection

Login as **PHLEBOTOMIST** MBSPHLEB001.

#	Action	Expected Result	✓
C1	Open the Collection Worklist for today/tomorrow (matching the collection date chosen in B3/B8).	Both Pathology orders from Section B appear, selectable (not greyed out) since they are fully paid — no "Payment pending" badge.	☐
C2	Select two same-order, same-specimen-type tests together and mark them collected as one batch draw.	Both items move to COLLECTED in one action; a single sample record is created.	☐
C3	Check the "Collected" tab for today.	Both just-drawn samples appear, regardless of their downstream receive status.	☐
C4	(Negative test) Try the collection worklist logged in as PATHOLOGIST (MBSPATH001) without any special tenant setting enabled.	403 Forbidden / page shows access denied — a Pathologist cannot collect samples by default.	☐

Section D — Lab Technician: Receive & Result Entry

Login as `LAB_TECHNICIAN` MBSLABTECH001.

#	Action	Expected Result	✓
D1	Open "Awaiting Receipt", find the sample collected in Section C, mark it Received.	Status moves COLLECTED → RECEIVED; sample now appears in the result-entry worklist.	☐
D2	Enter a normal-range result for the parameter (e.g. Hemoglobin within reference range).	Value saves; flag shows "Normal"; no critical alert raised.	☐
D3	For a second sample, enter a result at/below the configured critical-low threshold.	A Critical Alert fires automatically on submit — visible on the doctor's dashboard, not a manual step.	☐
D4	Submit both results for verification.	Items disappear from the Lab Technician's own queue — they cannot verify their own entries.	☐
D5	Test sample rejection: mark one collected-but-not-yet-verified sample as REJECTED (e.g. haemolysed).	Item flips back to ORDERED, a fresh collection task appears for the Phlebotomist, and — critically — no new bill line is created (check the original Bill's total is unchanged in Billing → Diagnostics).	☐

Section E — Pathologist: Verification & Release

Login as **PATHOLOGIST** MBSPATH001.

#	Action	Expected Result	✓
E1	Open the Verification worklist — confirm the two results from Section D are listed, including the critical one flagged.	Full context bundle visible per item (previous values, reference range, entering tech).	☐
E2	Acknowledge the critical alert raised in D3.	Alert moves from "open" to "acknowledged"; escalation timer stops.	☐
E3	Verify and release the normal-range result.	Status → VERIFIED → RELEASED; report becomes available; no new billing action needed (charge already posted at booking) — confirm in Billing that nothing duplicates.	☐
E4	(Negative test) As the same Pathologist, try to verify a result that this same login also entered (if applicable) or use a test tenant where Lab Technician and Pathologist happen to be the same login.	Blocked — "cannot verify your own entry" (BR-03), unless Separation of Duties has been explicitly turned off for this tenant (see Section K).	☐
E5	Open the released report and confirm the QR code / verification link.	Public verification page (no login) shows only report number, patient initials, date, genuine/not-genuine — no clinical content.	☐

Section F — Radiographer: Perform Imaging Study

Login as **RADIOGRAPHER** MBSRADIOG001.

#	Action	Expected Result	✓
F1	Open the Imaging worklist for the slot booked in Section B (B2/B3).	Appointment appears, selectable (payment already collected, no gate to fight).	☐
F2	If the test requires consent, try to check the patient in without ticking the consent checkbox.	Check-in is blocked until consent is confirmed.	☐
F3	Check in, perform the study, and draft initial findings/impression in the same step.	Study status moves to performed/drafted; draft is editable by the Radiologist next, not yet released.	☐

Section G — Radiologist: Author, Verify & Release

Login as **RADIOLOGIST** MBSRADIOL001, then MBSRADIOL002 for the separation-of-duties check.

#	Action	Expected Result	✓
G1	As MBSRADIOL001, open the draft from Section F, edit the findings/impression.	Edits save; `authoredById` recorded as MBSRADIOL001.	☐
G2	As the <i>same</i> MBSRADIOL001, attempt to verify & release this same study.	Blocked — same-person entered-and-verify check (BR-03 equivalent for Imaging).	☐
G3	Log out, log in as MBSRADIOL002, verify & release the same study.	Succeeds — different verifier; report releases; charge was already posted at booking, confirm no duplicate bill line appears.	☐
G4	(Negative test) As MBSRADIOL002, try the Pathology verification endpoint/worklist directly.	Rejected — a Radiologist cannot verify a Pathology item, and vice versa (domain cross-check).	☐

Section H — Outsourced Lab Workflow

Book an Outsourced test via Reception first (repeat Section B for an Outsourced-type test), then continue as **LAB_TECHNICIAN** and **PATHOLOGIST**.

#	Action	Expected Result	✓
H1	Book and collect an Outsourced test the same way as Pathology (same COLLECTION appointment type).	Payment mandatory at booking, same as any other type; item reaches RECEIVED status after collection+receipt.	☐
H2	As Lab Technician, batch this item with any other RECEIVED outsourced items bound for the same partner lab into one dispatch.	One dispatch note created with a dispatch number, courier, tracking ref; `expectedReturnAt` computed from the test's TAT.	☐
H3	If the dispatch has 2+ items, mark only one as returned.	Dispatch status becomes PARTIAL_RETURN; the other item stays DISPATCHED independently — nothing blocks on the slower item.	☐
H4	Enter the result for the returned item (as Lab Technician) and verify as Pathologist (different login).	Report releases with a "Tested at [Partner Lab]" footnote; hospital's own Pathologist still signs off before release.	☐
H5	Set the dispatch's expected-return date in the past (or use an already-overdue fixture) and check the dashboard.	Overdue badge surfaces before the patient has to complain.	☐

Section I — Billing & Cashier

Login as **CASHIER** MBSCA2026001 (or Reception/Admin).

#	Action	Expected Result	✓
I1	Open Billing, filter by the Diagnostics tab (previously labelled "Lab").	Every order booked in Sections B, H shows up here, each dated to its actual booking day — not the day it was later verified.	☐
I2	Open one bill with a mixed-type order (Pathology + Imaging from B2).	Both line items show, each individually tagged PATHOLOGY / RADIOLOGY — not lumped as one undifferentiated "Lab" charge.	☐
I3	Confirm the day's "Total Collected" / collection-rate figures include these diagnostics bills.	Numbers match — diagnostics revenue is no longer invisible until days later at verification.	☐

Section J — Admin: Catalogue & Resource Setup

Login as **ADMIN** ADMIN-001. Run this section first if the preconditions on the cover page aren't already met.

#	Action	Expected Result	✓
J1	Create a new Pathology test with one numeric parameter but no reference range, then try to activate it.	Blocked — a fallback (sex=ANY, all-ages) reference range row is required before activation (BR-10).	☐
J2	Create a new Imaging test with zero parameters and activate it.	Activates successfully — the BR-10 reference-range gate only applies to Pathology, not Imaging.	☐
J3	Create/edit an Imaging resource (machine or room): set working hours, slot duration, and a blackout date.	Saves; the blackout date correctly removes that day from the slot picker reception sees during booking.	☐
J4	Map the new Imaging test to the resource, and set a price for it on the default Cash rate list.	Test becomes bookable end-to-end (appears with a price in Reception's booking screen).	☐
J5	Fire two near-simultaneous bookings for the exact same resource/date/time (open two tabs, submit within a second of each other).	Exactly one succeeds; the other receives a clean 409 "slot just taken" — never a silent double-booking or a 500 error.	☐

Section K — Super Admin: Tenant Settings

Login to the Super Admin portal.

#	Action	Expected Result	✓
K1	Open the MBS tenant's Diagnostics settings. Confirm "Separation of Duties" shows Enforced (default) for both Pathology and Imaging.	Default-on for every tenant unless explicitly changed.	☐
K2	Turn OFF Separation of Duties for Pathology only, save.	Re-run E4 (self-verify) — a Pathologist can now verify their own entry for Pathology, but Imaging's rule is untouched.	☐
K3	Turn ON "Solo Staff Mode (Role Consolidation)" for Pathology.	Re-run C4 (negative test) — a Pathologist login can now reach the Collection worklist and the result-entry screen, in addition to Verification.	☐
K4	With both K2 and K3 enabled, log in as MBSPATH001 alone and run collect → enter result → verify → release for one order, start to finish, with no other login involved.	Full solo pipeline succeeds end to end — this is the intended "hospital with only one pathologist" scenario.	☐
K5	While Role Consolidation is ON, log in as MBSPHLEB001 (a junior role) and try to reach the Verification worklist.	Still blocked (403) — consolidation only ever widens access upward (senior → junior tasks), never downward.	☐
K6	Revert K2 and K3 back to the default (Enforced / Disabled) once testing is complete.	Tenant returns to standard multi-staff behavior.	☐

Sign-off

Section	Tested by	Date	Result (Pass / Fail / Blocked)	Notes
A — Doctor				
B — Reception (booking & payment)				
C — Phlebotomist				
D — Lab Technician				
E — Pathologist				
F — Radiographer				
G — Radiologist				
H — Outsourced				
I — Billing / Cashier				
J — Admin setup				
K — Super Admin settings				